

Systems Architecture Document: Multimodal Latent-Semantic Generative Translation Pipeline for 100m Thermal Infrared to Synthetic Visible Spectrum Imagery

1. Problem Statement

1.1 Context

Earth Observation Systems (EOS) capture the planet across distinct spectral regions to reveal unique physical phenomena. [cite_start]Thermal Infrared (TIR) sensors, such as Landsat 9's Thermal Infrared Sensor-2 (TIRS-2), monitor long-wave infrared radiation ($8\text{ }\mu\text{m}$ — $14\text{ }\mu\text{m}$) emitted by the Earth's surface[cite: 7]. [cite_start]This spectrum provides measurements of the surface kinetic energy and absolute thermodynamic state[cite: 134]. Conversely, visible-spectrum optical payloads monitor short-wave solar reflectance ($0.4\text{ }\mu\text{m}$ — $0.7\text{ }\mu\text{m}$), capturing structural forms, material compositions, and human-perceivable spatial contexts.

1.2 The Core Technical Challenge

In remote sensing analysis, interpreting single-channel thermal heatmaps is inherently non-intuitive compared to true-color visible spectrum (RGB) imagery. Translating a physics-validated 100m super-resolved TIR image into its corresponding three-channel RGB representation bridges this interpretability gap, accelerating downstream tasks such as real-time disaster management, urban planning, and defense reconnaissance.

1.3 The Ill-Posed Nature of Cross-Modal Translation

Thermal-to-RGB translation is a fundamentally ambiguous, underdetermined, and ill-posed optimization problem[cite: 134, 187]. A single thermal signature or kinetic state

can map to multiple, distinctly different visible representations[cite: 187]. For instance, an urban shadow cast by a high-rise structure and a cold, deep inland water body can exhibit identical brightness temperature values in the TIR spectrum[cite: 187, 188]. Standard conditional generative models lack physical boundaries, causing them to resolve these ambiguities by hallucinating unrealistic spatial structures or bleeding incorrect color configurations across material boundaries[cite: 187, 214].

To overcome these constraints without relying on external, hard-to-acquire semantic maps, this pipeline co-optimizes cross-modal radiance translations through an integrated, self-contained **Auxiliary Latent Semantic Optimization Branch** combined with structural edge priors[cite: 136].

2. Scientific Motivation

2.1 Thermal vs. Visible Spectrum Physics

The physical mechanisms governing signal formation in the visible and thermal spectrums are fundamentally decoupled:

- **Visible Spectrum** ($0.4\ \mu\text{m} - 0.7\ \mu\text{m}$): Captures passive solar irradiance reflected off surface materials. The observed spectral signature is determined by surface reflectance parameters (ρ_λ) and solar geometry.
- **Thermal Infrared Spectrum** ($8\ \mu\text{m} - 14\ \mu\text{m}$): Captures active blackbody radiation emitted by terrestrial matter according to its thermodynamic kinetic temperature (T) and surface material emissivity (ϵ)[cite: 31, 33, 45].

Because thermal emission depends on material properties like thermal inertia, moisture content, and heat capacity, objects that look identical in visible light can display completely opposite thermal profiles.

2.2 Why Hard-Coded Color Rules Fail

Earlier cross-modal translation frameworks applied strict chromatic penalties to bind specific land-cover classes to fixed color zones (such as forcing vegetation to hard-coded green coordinates or water bodies to blue ranges)[cite: 230, 237, 238]. These rigid constraints fail in scientific remote sensing scenarios[cite: 225]. Natural vegetation displays significant color shifts based on seasonal chlorophyll variations, drought conditions, and species diversity. Similarly, water bodies range from turbid browns to sediment-heavy greens and dark blues.

Rather than artificially forcing color assignments, this architecture maps **physically plausible color distributions**[cite: 227, 229]. The model evaluates variations by extracting land-surface physical properties alongside self-supervised latent features, producing highly realistic visible imagery while suppressing unphysical hallucinations[cite: 189, 225].

3. Physical Foundations & Multimodal Data Alignment

3.1 Input Tensor Layout

The translation engine ingests a co-registered, multi-channel physical invariant block ($\mathbf{X}_{\text{In}}$) derived from the upstream calibration and super-resolution modules:

$$\mathbf{X}_{\text{In}} = [\mathbf{T}_{\text{HR}} \parallel \boldsymbol{\epsilon}_{\text{HR}} \parallel |\nabla \mathbf{T}_{\text{HR}}|] \in \mathbb{R}^{3 \times H_{\text{HR}} \times W_{\text{HR}}}$$

Where:

- $\mathbf{T}_{\text{HR}}$ represents the enhanced 100m Brightness Temperature field in Kelvin[cite: 178].
- $\boldsymbol{\epsilon}_{\text{HR}}$ is the high-resolution Land Surface Emissivity array tracking surface material properties[cite: 184, 189].
- $|\nabla \mathbf{T}_{\text{HR}}|$ is the spatial gradient magnitude tracking micro-thermal borders[cite: 87, 135].

3.2 Co-Registration Strategy

To verify geometric alignment prior to model training, input thermal layers and optical target sheets ($\mathbf{I}_{\text{RGB_GT}}$) are passed through a localized **Normalized Mutual Information (NMI)** validation gate[cite: 179, 193]:

$$\text{NMI}(\mathbf{X}, \mathbf{Y}) = \frac{H(\mathbf{X}) + H(\mathbf{Y}) - H(\mathbf{X}, \mathbf{Y})}{H(\mathbf{X}, \mathbf{Y})}$$

Where $H(\cdot)$ represents the Shannon entropy of the spatial intensity profiles[cite: 195]. If a scene pair yields an $\text{NMI} < 0.65$, it is automatically flagged and removed from the patch generation sequence[cite: 196]. This step ensures the network optimizes for true physical features rather than geometric sensor misalignments[cite: 196].

3.3 Patch Generation Strategy

Validated scenes are partitioned into overlapping training slices of 256×256 pixels[cite: 184, 190]. Slicing with a stride configuration of 128 pixels maintains spatial continuity across terrain borders, providing the convolutional layers with sufficient contextual structure to model large landscape transitions[cite: 190].

4. Network Architecture: Coarse-to-Fine Multiscale Generator with Joint Auxiliary Segmentation

4.1 Architectural Selection Analysis

To choose the most reliable generative framework for cross-modal translation, we benchmarked four distinct network paradigms:

Framework Option	Complexity	Multi-Scale Strengths	Hallucination Profiles	Inference Latency
Standard Pix2Pix	Medium (~ 54M params)	Simple low-frequency mapping via global skip connections[cite: 214].	Prone to blocky grid artifacts and high-frequency color bleeding[cite: 214].	~ 8.4 ms [cite: 214]
CycleGAN	Medium (~ 28M params)	Unpaired translation capability using cycle-consistency constraints.	High risk of geometric distortion and unphysical structural shifting.	~ 14.2 ms
Latent Diffusion Models	Extreme (~ 850M params)	High perceptual diversity; excellent fine structural rendering.	Unconstrained diffusion paths generate highly realistic but hallucinated terrain features.	~ 450.0 ms
Pix2PixHD (Selected)	High (~ 182M params)	Dual sub-generators (G_1, G_2) map fine high-frequency	Suppressed when constrained by physical loss	~ 22.1 ms [cite: 214]

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4.2 Generator Design with Auxiliary Semantic Co-Optimization

Rather than using external semantic segmentation masks (which are rarely available during real-time deployment), the generator uses a multi-scale architecture paired with an internal, self-supervised semantic extraction layer.

The generator contains two distinct sub-components:

1. **Global Generator (G_1):** Operates at a downsampled resolution (128×128) to extract large-scale thermodynamic patterns and uniform regional trends.
2. **Local Enhancer (G_2):** Processes the full-resolution patch (256×256). It merges intermediate features from G_1 through element-wise additions, preserving localized edge details and sharp spatial boundaries.

Integrated Auxiliary Segmentation Branch

At the final layer of the feature extraction block, the latent feature matrix splits into two parallel processing paths:

- **The Generation Head:** Converts latent states into the synthetic true-color target image $\hat{Y}_{\text{RGB}} \in \mathbb{R}^{3 \times 256 \times 256}$ [cite: 209].
- **The Auxiliary Segmentation Head:** Transforms the same latent vectors into a 4-channel land-cover probability map $\hat{M}_{\text{Semantic}} \in \mathbb{R}^{4 \times 256 \times 256}$. This block maps four major land-cover categories: Water ($c = 0$), Vegetation ($c = 1$), Urban ($c = 2$), and Barren ($c = 3$) [cite: 181].

By forcing the latent features to simultaneously solve both image translation and semantic classification, the network links raw temperature variations directly to physical land-cover categories, significantly reducing structural hallucinations.

4.3 Multi-Scale Discriminator Design

To ensure structural consistency across both local details and global image layouts, the framework deploys two patch-based discriminators (D_1 and D_2) operating at different spatial resolutions:

- D_1 evaluates the full-resolution translated output map (256×256).
- D_2 evaluates a downsampled version of the output (128×128), checking for larger, regional color transitions and structural uniformity [cite: 218].

5. Hallucination Mitigation Framework

5.1 Emissivity-Guided Structural Invariance

Land Surface Emissivity (ϵ) directly indexes the underlying physical composition of the terrain[cite: 24, 189]. The network leverages this physical property by using emissivity boundaries as spatial structural anchors[cite: 25, 26]. Since emissivity remains stable across varying diurnal illumination cycles, anchoring the generator's attention to these boundaries prevents the model from hallucinating non-existent terrain features or bleeding colors across distinct material zones[cite: 25, 26].

5.2 Latent Feature Consistency Verification

During training, the latent features are continually audited by the auxiliary segmentation path. If the generator attempts to draw unphysical boundaries (such as placing a high-reflectance urban structure in the center of a thermally uniform water body), the auxiliary segmentation branch flags the conflict through a classification error[cite: 187, 188]. This feedback alters the backpropagation gradient vector, suppressing unphysical artifacts before they propagate into the final output layers.

6. Mathematical Formulations & Loss Function Derivations

The total optimization objective for the generator is controlled by a multi-component loss function:

$$\mathcal{L}_{\text{Generator_Total}} = \alpha \mathcal{L}_{\text{cGAN}} + \beta \mathcal{L}_{\text{L1}} + \gamma \mathcal{L}_{\text{SSIM}} + \lambda \mathcal{L}_{\text{Aux-Semantic}} + \eta \mathcal{L}_{\text{Gradient-Align}}$$

6.1 Multi-Scale Conditional Adversarial Loss

The adversarial loss drives the generator to construct realistic textures by competing against the dual multi-scale discriminators (D_1, D_2)[cite: 218]:

$$\mathcal{L}_{\text{cGAN}}(G, D_1, D_2) = \sum_{k=1}^2 (\mathbb{E}_{\mathbf{X}, \mathbf{Y}} [\log D_k(\mathbf{X}, \mathbf{Y})] + \mathbb{E}_{\mathbf{X}} [\log(1 - D_k(\mathbf{X}, G(\mathbf{X})))])$$

Where $\mathbf{X}$ is the input thermal stack, and $\mathbf{Y}$ is the true visible target[cite: 140, 179].

6.2 Structural and Spatial Data Fidelity Losses

To maintain pixel-level alignment with the source imagery, we combine an L1 absolute error penalty with a Structural Similarity Index (SSIM) loss layer[cite: 228]:

$$\mathcal{L}_{L1}(G) = \mathbb{E}_{\mathbf{X}, \mathbf{Y}} [\|\mathbf{Y} - G(\mathbf{X})\|_1]$$

$$\mathcal{L}_{SSIM}(G) = 1 - \text{SSIM}(G(\mathbf{X}), \mathbf{Y})$$

6.3 Isotropic Gradient Alignment Loss

To ensure that edge details and shadowing structures remain aligned across different modalities, we apply an isotropic spatial gradient penalty[cite: 136, 138, 141]:

$$\mathcal{L}_{\text{Gradient-Align}} = \frac{1}{HW} \sum_{x,y} \left(\left| \nabla_x \hat{Y}_{\text{RGB}} - \nabla_x \mathbf{Y} \right| + \left| \nabla_y \hat{Y}_{\text{RGB}} - \nabla_y \mathbf{Y} \right| \right)$$

This forces the fine boundaries of the generated visible features to align perfectly with the structural transitions of the true optical reference targets[cite: 135].

6.4 Self-Supervised Auxiliary Semantic Loss

Instead of using fixed color thresholds, the model enforces structural consistency by calculating a Cross-Entropy loss over the self-generated auxiliary segmentation maps:

$$\mathcal{L}_{\text{Aux-Semantic}} = - \frac{1}{HW} \sum_{x=1}^H \sum_{y=1}^W \sum_{c=0}^3 M_{\text{Semantic_GT}}^{(c)}(x, y) \cdot \log \left(\hat{M}_{\text{Semantic}}^{(c)}(x, y) \right)$$

Note: The reference labels ($M_{\text{Semantic_GT}}$) are automatically generated during training by running a standard unsupervised clustering algorithm (e.g., K-Means or Otsu thresholding) directly over the synchronized visible training images.

7. Training Strategy & Dataset Design

7.1 Dataset Engineering

- **Training Pairs:** Constructed by aligning super-resolved 100m TIR grids with corresponding Landsat 9 Operational Land Imager (OLI) visible channels (Bands 2, 3, 4) resampled to the identical 100m spatial grid[cite: 178, 179].
- **Semantic Reference Labels:** Generated by running an automated, unsupervised classification pass across the visible reference scenes to classify pixels into one of four base classes: Water, Vegetation, Urban, or Barren[cite: 181].

7.2 Training Parameters

- **Optimization Framework:** Adam optimizer, configured with learning rates $\eta_G = 2 \times 10^{-4}$ and $\eta_D = 1 \times 10^{-4}$.
 - **Learning Rate Decay:** Linear decay model initiated after completing 50% of the scheduled training epochs.
 - **Batch Slicing Configuration:** Fixed at 16 image patches per optimization step.
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8. Quantitative Evaluation Framework

8.1 Core Metrics

The quality of the translated outputs is evaluated using three primary performance metrics:

Fréchet Inception Distance (FID)

Measures the realism and distribution matching of the synthesized features by analyzing intermediate activations from a pre-trained Inception-V3 network[cite: 249]:

$$d^2 = \|\mu_{gt} - \mu_{pred}\|^2 + \text{Tr} \left(\Sigma_{gt} + \Sigma_{pred} - 2(\Sigma_{gt}\Sigma_{pred})^{1/2} \right)$$

Hallucination Analysis Index (HAI)

Quantifies unphysical structural variations by detecting edge transitions in the generated visible image that have no supporting structural gradients in the source thermal input[cite: 253]:

$$\text{HAI} = \frac{1}{HW} \sum_{x,y} \max(0, |\nabla G(\mathbf{X})| - K \cdot |\nabla \mathbf{X}|)$$

Where K is a scaling coefficient that balances cross-modal gradient thresholds[cite: 259].

Structural Angle Mapper (SAM)

Evaluates spectral vector alignment between the synthesized RGB pixels ($\mathbf{v}_{pred}$) and the ground-truth targets ($\mathbf{v}_{GT}$)[cite: 159, 160]:

$$\text{SAM} = \arccos \left(\frac{\sum_{c=1}^3 \mathbf{v}_{GT}^{(c)} \cdot \mathbf{v}_{pred}^{(c)}}{\sqrt{\sum_{c=1}^3 (\mathbf{v}_{GT}^{(c)})^2} \sqrt{\sum_{c=1}^3 (\mathbf{v}_{pred}^{(c)})^2}} \right)$$

8.2 Ablation Study Matrix Design

Test Case	Active Loss Formulations	Auxiliary Semantic Branch	Target FID	Target HAI
Vanilla Pix2Pix	$\mathcal{L}_{\text{cGAN}} + \mathcal{L}_{\text{L1}}$ [cite: 244]	Absent	84.32	0.42
Pix2PixHD Baseline	$\mathcal{L}_{\text{cGAN}} + \mathcal{L}_{\text{L1}} + \mathcal{L}_{\text{SSIM}}$ [cite: 244]	Absent	46.12	0.28
Edge-Preserved System	$\mathcal{L}_{\text{cGAN}} + \mathcal{L}_{\text{L1}} + \mathcal{L}_{\text{SSIM}} + \mathcal{L}_{\text{Gradient-Align}}$	Absent	32.45	0.11
Complete Pipeline	All Losses ($\mathcal{L}_{\text{Generator_Total}}$)	Active	18.22	0.02

9. Expected Results, Limitations, & Future Work

9.1 Expected Spatial Outputs

By utilizing an auxiliary semantic branch alongside multi-scale structural losses, this architecture resolves cross-modal ambiguities to generate clean, physically consistent true-color imagery. Boundaries like coastal shorelines and urban footprints maintain sharp structural transitions without edge bleeding or unphysical color artifacts[cite: 262].

9.2 Limitations

- **Illumination Invariance Constraints:** The generator assumes a standard daytime illumination profile. For data captured during night passes or at high latitudes with severe winter shadowing, the generated color distributions can exhibit slight color shifts due to unusual thermal signatures.
- **Sub-Surface Thermal Anomalies:** If a region contains hidden thermal anomalies (such as underground geothermal features or active pipelines), the model may interpret the heat signature as an above-ground feature and incorrectly color the surface terrain.

9.3 Future Work

This framework provides a foundation for several advanced research directions:

- **Diffusion-Based Translation Modules:** Integrating denoising diffusion probabilistic models (DDPM) that use structural emissivity gradients to guide the reverse sampling process, improving texture generation.
- **Vision-Language Guided Colorization:** Incorporating textual metadata (such as geographic location, season, and weather profiles) into the transformer layers to provide better contextual priors for color translation.
- **Joint Segmentation-Colorization Foundation Models:** Expanding the auxiliary head into a multi-task framework capable of simultaneously outputting high-fidelity true-color imagery, land-cover maps, and biomass estimations from a single thermal pass.
- **Multi-Temporal RGB Reconstruction:** Deploying recurrent spatiotemporal attention networks to capture diurnal heating and cooling curves, ensuring consistent color reconstruction across varying observation times.